Calculus II
Homework
Integration by parts

1. Let $x \in (0,1)$. Express the following using x and $\sqrt{1-x^2}$.

   (a) $\sin(\arcsin(x))$.

   answer: x

   (b) $\sin(2\arcsin(x))$.

   answer: $2x\sqrt{1-x^2}$

   (c) $\sin(3\arcsin(x))$.

   answer: $-4x^3+3x$

   (d) $\sin(\arccos(x))$.

   answer: $\sqrt{1-x^2}$

   (e) $\sin(2\arccos(x))$.

   answer: $2x\sqrt{1-x^2}$

   (f) $\sin(3\arccos(x))$.

   answer: $\begin{aligned}&\left(4x^2-1\right)\sqrt{1-x^2}\\ &=-4\left(\sqrt{1-x^2}\right)^3+3\sqrt{1-x^2}\end{aligned}$

   (g) $\cos(2\arcsin(x))$.

   answer: $1-2x^2$

   (h) $\cos(3\arccos(x))$.

   answer: $4x^3-3x$

Solution. 1.b. Let $y = \arcsin x$. Then $\sin y = x$, and we can draw a right triangle with opposite side length x and hypotenuse length 1 to find the other trigonometric ratios of y.

Then $\cos y = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$. Now we use the double angle formula to find $\sin(2\arcsin x)$.

$$\begin{aligned}\sin(2\arcsin x) &= \sin(2y)\\ &= 2\sin y\cos y\\ &= 2x\sqrt{1-x^2}.\end{aligned}$$

Solution. 1.c. Use the result of Problem 1.b. This also requires the addition formula for sine:

$$\sin(A+B) = \sin A\cos B + \sin B\cos A,$$

and the double angle formula for cosine:

$$\cos(2y) = \cos^2 y - \sin^2 y.$$

$$\begin{aligned}\sin(3\arcsin x) &= \sin(3y)\\ &= \sin(2y+y)\\ &= \sin(2y)\cos y + \sin y\cos(2y) && \text{Use addition formula}\\ &= (2\sin y\cos y)\cos y + \sin y(\cos^2 y - \sin^2 y) && \text{Use double angle formulas}\\ &= 2\sin y\cos^2 y + \sin y\cos^2 y - \sin^3 y\\ &= 3\sin y\cos^2 y - \sin^3 y\\ &= 3\sin y(1-\sin^2 y) - \sin^3 y\\ &= 3x(1-x^2) - x^3\\ &= 3x - 4x^3.\end{aligned}$$

The solution is complete. A careful look at the solution above reveals a strategy useful for problems similar to this one.

(a) Identify the inverse trigonometric expression- $\arcsin x, \arccos x, \arctan x, \ldots$. In the present problem that was $y = \arcsin x$.

(b) The problem is therefore a trigonometric function of y.

(c) Using trig identities and algebra, rewrite the problem as a trigonometric expression involving only the trig function that transforms y to x. In the present problem we rewrote everything using $\sin y$.

(d) Use the fact that $\sin(\arcsin x) = x$, $\cos(\arccos x) = x$, ..., etc. to simplify.

Solution. 1.f We use the same strategy outlined in the end of the solution of Problem 1.c. Set $y = \arccos x$ and so $\cos(y) = x$. Therefore:

$$\begin{aligned}\sin(3y) &= \sin(2y+y)\\ &= \sin(2y)\cos y + \sin y\cos(2y)\\ &= 2\sin y\cos y\cos y + \sin y(2\cos^2 y - 1)\\ &= 2\sin y\cos^2 y + \sin y(2\cos^2 y - 1)\\ &= \sin y(4\cos^2 y - 1) && \text{use } \cos y = x,\ \sin y = \sqrt{1-x^2}\\ &= \sqrt{1-x^2}(4x^2-1) \quad .\end{aligned}$$

2. Express as the following as an algebraic expression of x. In other words, "get rid" of the trigonometric and inverse trigonometric expressions.

(a) $\cos^2(\arctan x)$. answer: $\frac{1}{1+x^2}$

(b) $-\sin^2(\operatorname{arccot} x)$. answer: $-\frac{1}{1+x^2}$

(c) $\dfrac{1}{\cos(\arcsin x)}$. answer: $\frac{1}{\sqrt{1-x^2}}$

(d) $-\dfrac{1}{\sin(\arccos x)}$. answer: $-\frac{1}{\sqrt{1-x^2}}$

Solution. 2.b. We follow the strategy outlined in the end of the solution of Problem 1.c. We set $y = \operatorname{arccot} x$. Then we need to express $-\sin^2 y$ via $\cot y$. That is a matter of algebra:

$$\begin{aligned} -\sin^2(\operatorname{arccot} x) &= -\sin^2 y && \text{Set } y = \operatorname{arccot} x \\ &= -\frac{\sin^2 y}{\sin^2 y + \cos^2 y} && \text{use } \sin^2 y + \cos^2 y = 1 \\ &= -\frac{1}{\frac{\sin^2 y + \cos^2 y}{\sin^2 y}} \\ &= -\frac{1}{1 + \cot^2 y} && \text{Substitute back } \cot y = x \\ &= -\frac{1}{1 + x^2} \quad . \end{aligned}$$

3. Rewrite as a rational function of t. This problem will be later used to derive the Euler substitutions (an important technique for integrating).

(a) $\cos(2\arctan t)$.

answer: $\frac{1-t^2}{1+t^2}$.

(b) $\sin(2\arctan t)$.

answer: $\frac{2t}{1+t^2}$

(c) $\tan(2\arctan t)$.

answer: $\frac{2t}{1-t^2}$

(d) $\cot(2\arctan t)$.

answer: $\frac{1}{2}\left(\frac{1}{t} - t\right)$.

(e) $\csc(2\arctan t)$.

answer: $\frac{1}{2}\left(t + \frac{1}{t}\right)$

(f) $\sec(2\arctan t)$.

answer: $\frac{1+t^2}{1-t^2}$

(g) $\cos(2\operatorname{arccot} t)$.

answer: $\frac{t^2-1}{t^2+1}$

(h) $\sin(2\operatorname{arccot} t)$.

answer: $\frac{2t}{t^2+1}$

(i) $\tan(2\operatorname{arccot} t)$.

answer: $\frac{2t}{t^2-1}$

(j) $\cot(2\operatorname{arccot} t)$.

answer: $\frac{1}{2}\left(t - \frac{1}{t}\right)$

(k) $\csc(2\operatorname{arccot} t)$.

answer: $\frac{1}{2}\left(t + \frac{1}{t}\right)$

(l) $\sec(2\operatorname{arccot} t)$.

answer: $\frac{t^2+1}{t^2-1}$

Solution. 3.a Set $z = \arctan t$, and so $\tan z = t$. Then

$$\begin{aligned} \cos(2\arctan t) &= \cos(2z) \\ &= \frac{\cos(2z)}{1} && \text{use double angle formulas} \\ &= \frac{\cos^2 z - \sin^2 z}{\cos^2 z + \sin^2 z} && \text{and } 1 = \sin^2 z + \cos^2 z \\ &= \frac{(\cos^2 z - \sin^2 z)\frac{1}{\cos^2 z}}{(\sin^2 z + \cos^2 z)\frac{1}{\cos^2 z}} && \text{divide top and bottom by } \cos^2 z \\ &= \frac{1 - \tan^2 z}{1 + \tan^2 z} \\ &= \frac{1 - t^2}{1 + t^2} \quad . \end{aligned}$$

Solution. 3.d Set $z = \arctan t$, and so $\tan z = t$. Then

$$\begin{aligned} \cot(2\arctan t) &= \cot(2z) \\ &= \frac{\cos(2z)}{\sin(2z)} && \text{use double angle formulas} \\ &= \frac{\cos^2 z - \sin^2 z}{2\sin z \cos z} \\ &= \frac{1 - \tan^2 z}{2\tan z} \\ &= \frac{1 - t^2}{2t} \\ &= \frac{1}{2}\left(\frac{1}{t} - t\right) \quad . \end{aligned}$$

4. Compute the derivative (derive the formula).

   (a) $(\arctan x)'$. answer: $\frac{1}{1+x^2}$

   (b) $(\operatorname{arccot} x)'$. answer: $-\frac{1}{1+x^2}$

   (c) $(\arcsin x)'$. answer: $\frac{1}{\sqrt{1-x^2}}$

   (d) $(\arccos x)'$. answer: $-\frac{1}{\sqrt{1-x^2}}$

   (e) Let arcsec denote the inverse of the secant function. Compute $(\operatorname{arcsec} x)'$. answer: $\frac{1}{x\sqrt{x^2-1}}$

5. (a) Let $a+b \neq k\pi$, $a \neq k\pi + \frac{\pi}{2}$ and $b \neq k\pi + \frac{\pi}{2}$ for any $k \in \mathbb{Z}$ (integers). Prove that

$$\frac{\tan a + \tan b}{1 - \tan a \tan b} = \tan(a+b) \quad .$$

   (b) Let x and y be real. Prove that, for $xy \neq 1$, we have

$$\arctan x + \arctan y = \arctan\left(\frac{x+y}{1-xy}\right)$$

   if the left hand side lies between $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Solution. 5.a We start by recalling the formulas

$$\begin{aligned} \cos(a+b) &= \cos a \cos b - \sin a \sin b \\ \sin(a+b) &= \sin a \cos b + \sin b \cos a \quad . \end{aligned}$$

These formulas have been previously studied; alternatively they follow from Euler's formula and the computation

$$\begin{aligned} \cos(a+b) + i\sin(a+b) &= e^{i(a+b)} = e^{ia}e^{ib} = (\cos a + i \sin a)(\cos b + i \sin b) \\ &= \cos a \cos b - \sin a \sin b + i(\sin a \cos b + \sin b \cos a) \quad . \end{aligned}$$

Now 5.a is done via a straightforward computation:

$$\begin{aligned} \tan(a+b) &= \frac{\sin(a+b)}{\cos(a+b)} = \frac{\sin a \cos b + \sin b \cos a}{\cos a \cos b - \sin a \sin b} = \frac{(\sin a \cos b + \sin b \cos a)\frac{1}{\cos a \cos b}}{(\cos a \cos b - \sin a \sin b)\frac{1}{\cos a \cos b}} \\ &= \frac{\tan a + \tan b}{1 - \tan a \tan b} \quad . \end{aligned} \tag{1}$$

5.b is a consequence of 5.a. Let $a = \arctan x$, $b = \arctan y$. Then (**??**) becomes

$$\tan(\arctan x + \arctan y) = \frac{\tan(\arctan x) + \tan(\arctan y)}{1 - \tan(\arctan x)\tan(\arctan y)} = \frac{x+y}{1-xy} \quad ,$$

where we use the fact that $\tan(\arctan w) = w$ for all w. We recall that $\arctan(\tan z) = z$ whenever $z \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Now take $\arctan$ on both sides of the above equality to obtain

$$\arctan x + \arctan y = \arctan\left(\frac{x+y}{1-xy}\right) \quad .$$

6. Evaluate the indefinite integral. Illustrate the steps of your solutions.

   (a) $\int x \sin x \mathrm{d}x$. answer: $-x\cos x + \sin x + C$

   (b) $\int xe^{-x}\mathrm{d}x$. answer: $-(1+x)e^{-x} + C$

   (c) $\int x^2 e^x \mathrm{d}x$. answer: $x^2e^x - 2xe^x + 2e^x + C$

   (d) $\int x\sin(-2x)\mathrm{d}x$. answer: $\frac{x}{2}\cos(-2x) + \frac{1}{4}\sin(-2x) + C$

   (e) $\int x^2\cos(3x)\mathrm{d}x$. answer: $\frac{x^2}{3}\sin(3x) + \frac{2x}{9}\cos(3x) - \frac{2}{27}\sin(3x) + C$

   (f) $\int x^2 e^{-2x}\mathrm{d}x$. answer: $-\frac{x^2e^{-2x}}{2} - \frac{xe^{-2x}}{2} - \frac{e^{-2x}}{4} + C$

   (g) $\int x\sin(2x)\mathrm{d}x$. answer: $-\frac{x}{2}\cos(2x) + \frac{1}{4}\sin(2x) + C$

   (h) $\int x\cos(3x)\mathrm{d}x$.

(i) $\int x^2 e^{2x} \mathrm{d}x$.

answer: $\frac{x^2}{2} e^{2x} - \frac{x}{2} e^{2x} + \frac{e^{2x}}{4} + C$

(j) $\int x^3 e^x \mathrm{d}x$.

answer: $x^3 e^x - 3x^2 e^x + 6xe^x - 6e^x + C$

Solution. 6.a.

$$\int x \underbrace{\sin x \mathrm{d}x}_{=\mathrm{d}(-\cos x)} = -\int x \mathrm{d}(\cos x) = -x\cos x + \int \cos x \mathrm{d}x = -x\cos x + \sin x + C \quad .$$

Solution. 6.c.

$$\begin{aligned} \int x^2 \underbrace{e^x \mathrm{d}x}_{\mathrm{d}(e^x)} &= \int x^2 \mathrm{d}e^x = x^2 e^x - \int e^x 2x \mathrm{d}x = x^2 e^x - \int 2x \mathrm{d}e^x \\ &= x^2 e^x - 2xe^x + \int 2e^x \mathrm{d}x = x^2 e^x - 2xe^x + 2e^x + C \quad . \end{aligned}$$

Solution. 6.f.

$$\begin{aligned} \int x^2 e^{-2x} \mathrm{d}x &= \int x^2 \mathrm{d}\left(\frac{e^{-2x}}{-2}\right) && \Big| \text{ Integrate by parts} \\ &= -\frac{x^2 e^{-2x}}{2} - \int \left(\frac{e^{-2x}}{-2}\right) \mathrm{d}\left(x^2\right) \\ &= -\frac{x^2 e^{-2x}}{2} + \int x e^{-2x} \mathrm{d}x \\ &= -\frac{x^2 e^{-2x}}{2} + \int x \mathrm{d}\left(\frac{e^{-2x}}{-2}\right) && \Big| \text{ Integrate by parts} \\ &= -\frac{x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} + \frac{1}{2}\int e^{-2x} \mathrm{d}x \\ &= -\frac{x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} + C \quad . \end{aligned}$$

7. Evaluate the indefinite integral. Illustrate the steps of your solutions.

(a) $\int x^2 \cos(2x) \mathrm{d}x$.

answer: $\frac{1}{2}x^2 \sin(2x) + \frac{1}{2}x\cos(2x) - \frac{1}{4}\sin(2x) + C$

(b) $\int x^2 e^{ax} \mathrm{d}x$, where a is a constant.

answer: $\frac{1}{a}x^2 e^{ax} - \frac{2}{a^2} x e^{ax} + \frac{2}{a^3} e^{ax} + C$

(c) $\int x^2 e^{-ax} \mathrm{d}x$, where a is a constant.

answer: $-\frac{1}{a}x^2 e^{-ax} - \frac{2}{a^2} x e^{-ax} - \frac{2}{a^3} e^{-ax} + C$

(d) $\int x^2 \frac{\left(e^{ax} + e^{-ax}\right)^2}{4} \mathrm{d}x$, where a is a constant.

answer: $\frac{1}{8}\left(a^{-1}x^2 e^{2ax} - a^{-1}x^2 e^{-2ax} - a^{-2}xe^{2ax} - a^{-2}xe^{-2ax} + \frac{1}{2}a^{-3}e^{2ax} - \frac{1}{2}a^{-3}e^{-2ax} + \frac{2}{3}x^3\right) + C$

(e) $\int \frac{1}{\cos^2 x} \mathrm{d}x$. (Hint: This problem does not require integration by parts. What is the derivative of $\tan x$?)

answer: $\tan x + C$

(f) $\int (\tan^2 x) \mathrm{d}x$. (Hint: This problem does not require integration by parts. We can use $\tan^2 x = \frac{1}{\cos^2 x} - 1$ and the previous problem.)

(g) $\int x \tan^2 x \mathrm{d}x$. (Hint: $\tan^2 x \mathrm{d}x = \mathrm{d}(F(x))$, where $F(x)$ is the answer from the preceding problem).

answer: $-\frac{x^2}{2} + x\tan x + \ln|\cos x| + C$

(h) $\int e^{-\sqrt{x}} \mathrm{d}x$.

answer: $-2\sqrt{x}e^{-\sqrt{x}} - 2e^{-\sqrt{x}} + C$

(i) $\int \cos^2 x \, \mathrm{d}x$.

answer: $\frac{1}{4}\sin(2x) + \frac{x}{2} + C$

(j) $\int \frac{x}{1+x^2} \mathrm{d}x$ (Hint: use substitution rule, don't use integration by parts)

answer: $\frac{\ln\left(1+x^2\right)}{2} + C$

(k) $\int (\arctan x) \mathrm{d}x$.

answer: $x\arctan x - \frac{\ln\left(1+x^2\right)}{2} + C$

(l) $\int (\arcsin x) \mathrm{d}x$.

answer: $x\arcsin x + \sqrt{1-x^2} + C$

(m) $\int (\arcsin x)^2 \mathrm{d}x$. (Hint: Try substituting $x = \sin y$.)

answer: $x(\arcsin x)^2 + 2\sqrt{1-x^2}\arcsin x - 2x + C$

(n) $\int \arctan\left(\frac{1}{x}\right) \mathrm{d}x$.

(o) $\int \sin x e^x \mathrm{d}x$

answer: $\frac{1}{2}(e^x \sin x - e^x \cos x) + C$

(p) $\int \cos x e^x \mathrm{d}x$

answer: $\frac{1}{2}(e^x \cos x + e^x \sin x) + C$

(q) $\int \sin(\ln(x))\mathrm{d}x.$

answer: $\frac{x}{2}(\sin(\ln x) - \cos(\ln x)) + C$

(r) $\int \cos(\ln(x))\mathrm{d}x.$

answer: $\frac{x}{2}(\cos(\ln x) + \sin(\ln x)) + C$

(s) $\int \ln x \mathrm{d}x$

answer: $x \ln|x| - x + C$

(t) $\int x \ln x \, \mathrm{d}x.$

answer: $\frac{1}{2}x^2 \ln|x| - \frac{x^2}{4} + C$

(u) $\int \frac{\ln x}{\sqrt{x}}\mathrm{d}x.$

answer: $2\sqrt{x}(\ln x - 2) + C$

(v) $\int (\ln x)^2 \mathrm{d}x.$

answer: $x(\ln x)^2 - 2x \ln x + 2x + C$

(w) $\int (\ln x)^3 \mathrm{d}x.$

answer: $x(\ln x)^3 - 3x(\ln x)^2 + 6x \ln x - 6x + C$

(x) $\int x^2 \cos^2 x \mathrm{d}x.$ (This problem is related to Problem 7.d as $\cos x = \frac{e^{ix}+e^{-ix}}{2}$).

answer: $\frac{1}{4}x^2 \sin(2x) + \frac{1}{4}x\cos(2x) - \frac{1}{8}\sin(2x) + \frac{1}{6}x^3$

Solution. 7.g.

$$\begin{array}{rcll}
\int x\tan^2 x\mathrm{d}x & = & \int x\left(\sec^2 x - 1\right)\mathrm{d}x & \left| \text{ use } \sec^2 x - 1 = \tan^2 x \right. \\
& = & \int x\left(\sec^2 x - 1\right)\mathrm{d}x & \\
& = & -\int x\mathrm{d}x + \int x\sec^2 x\mathrm{d}x & \left| \text{ use } \mathrm{d}(\tan x) = \sec^2 x\mathrm{d}x \right. \\
& = & -\frac{x^2}{2} + \int x\mathrm{d}(\tan x) & \left| \text{ integrate by parts} \right. \\
& = & -\frac{x^2}{2} + x\tan x - \int \tan x\mathrm{d}x & \\
& = & -\frac{x^2}{2} + x\tan x - \int \frac{\sin x}{\cos x}\mathrm{d}x & \left| \text{ use } \sin x\mathrm{d}x = -\mathrm{d}(\cos x) \right. \\
& = & -\frac{x^2}{2} + x\tan x + \int \frac{\mathrm{d}(\cos x)}{\cos x} & \left| \text{ Set } y = \cos x \right. \\
& = & -\frac{x^2}{2} + x\tan x + \int \frac{1}{y}\mathrm{d}y & \\
& = & -\frac{x^2}{2} + x\tan x + \ln|y| + C & \left| \text{ Substitute back } y = \cos x \right. \\
& = & -\frac{x^2}{2} + x\tan x + \ln|\cos x| + C \quad . &
\end{array}$$

Solution. 7.h.

$$\begin{array}{rcll}
\int e^{-\sqrt{x}}\mathrm{d}x & = & \int 2ye^{-y}\mathrm{d}y & \left| \begin{array}{rcl} \sqrt{x} & = & y \\ \frac{1}{2\sqrt{x}}\mathrm{d}x & = & \mathrm{d}y \\ \mathrm{d}x & = & 2y\mathrm{d}y \end{array} \right. \text{Subst.:} \\
& = & \int 2y\mathrm{d}\left(-e^{-y}\right) & \left| \text{ int. by parts} \right. \\
& = & -2ye^{-y} + 2\int e^{-y}\mathrm{d}y & \\
& = & -2ye^{-y} - 2e^{-y} + C & \\
& = & -2\sqrt{x}e^{-\sqrt{x}} - 2e^{-\sqrt{x}} + C \quad . &
\end{array}$$

Solution. 7.i. Later, we shall study general methods for solving trigonometric integrals that will cover this example. Let us however show one way to solve this integral by integration by parts.

$$
\begin{aligned}
\int \cos^2 x \mathrm{d}x =& x\cos^2 x - \int x \mathrm{d}(\cos^2 x) \\
=& x\cos^2 x - \int x 2\cos x(-\sin x)\mathrm{d}x \qquad \Big| \ \sin(2x) = 2\sin x\cos x \\
=& x\cos^2 x + \int x\sin(2x)\mathrm{d}x \\
=& x\cos^2 x + \int x \mathrm{d}\left(\frac{-\cos(2x)}{2}\right) \\
=& x\cos^2 x + x\left(\frac{-\cos(2x)}{2}\right) - \int \left(\frac{-\cos(2x)}{2}\right)\mathrm{d}x \\
=& \frac{x}{2}\left(2\cos^2 x - \cos(2x)\right) + \frac{\sin(2x)}{4} + C \qquad \Big| \ \cos(2x) = \cos^2 x - \sin^2 x \\
=& \frac{x}{2}\left(2\cos^2 x - (\cos^2 x - \sin^2 x)\right) + \frac{\sin(2x)}{4} + C \qquad \Big| \ \cos^2 x + \sin^2 x = 1 \\
=& \frac{x}{2} + \frac{\sin(2x)}{4} + C \quad .
\end{aligned}
$$

Solution. 7.k

$$
\begin{aligned}
\int \arctan x \mathrm{d}x &= x\arctan x - \int x \mathrm{d}(\arctan x) \\
&= x\arctan x - \int \frac{x}{x^2+1}\mathrm{d}x \\
&= x\arctan x - \int \frac{\frac{1}{2}\mathrm{d}(x^2)}{x^2+1} \\
&= x\arctan x - \int \frac{\frac{1}{2}\mathrm{d}(x^2+1)}{x^2+1} \\
&= x\arctan x - \frac{1}{2}\ln(x^2+1) + C \quad .
\end{aligned}
$$

Solution. 7.m.

$$
\begin{aligned}
\int (\arcsin x)^2 \mathrm{d}x &= \int \left(\arcsin(\sin y)\right)^2 \mathrm{d}(\sin y) \qquad \Big| \ \text{Set } x = \sin y \\
&= \int y^2 \cos y \mathrm{d}y = \int y^2 \mathrm{d}(\sin y) \qquad \Big| \ \text{Integrate by parts} \\
&= y^2\sin y - \int 2y\sin y \mathrm{d}y \\
&= y^2\sin y + \int 2y \mathrm{d}(\cos y) \qquad \Big| \ \text{Integrate by parts} \\
&= y^2\sin y + 2y\cos y - 2\int \cos y \mathrm{d}y \\
&= y^2\sin y + 2y\cos y - 2\sin y + C \qquad \Big| \ \text{Substitute } y = \arcsin x \\
&= x(\arcsin x)^2 \\
&\quad +2\sqrt{1-x^2}\arcsin x - 2x + C \quad .
\end{aligned}
$$

Solution. 7.o

$$
\begin{aligned}
\int \sin x \underbrace{e^x \mathrm{d}x}_{=\mathrm{d}e^x} &= \sin x e^x - \int e^x \mathrm{d}(\sin x) = \sin x e^x - \int \cos x \underbrace{e^x \mathrm{d}x}_{=\mathrm{d}e^x} \\
&= \sin x e^x - e^x\cos x + \int e^x \mathrm{d}(\cos x) \\
&= e^x\sin x - e^x\cos x - \int e^x \sin x \mathrm{d}x \qquad \Big| \ \begin{array}{l}\text{add } \int e^x\sin x\mathrm{d}x \\ \text{to both sides}\end{array} \\
2\int \sin x e^x \mathrm{d}x &= \sin x e^x - e^x\cos x \\
\int \sin x e^x \mathrm{d}x &= \frac{1}{2}\left(\sin x e^x - e^x\cos x\right) \quad .
\end{aligned}
$$

Solution. 7.q.

$$
\begin{aligned}
\int \sin(\ln x)\mathrm{d}x &= x\sin(\ln x) - \int x\mathrm{d}(\sin(\ln x)) && \text{int. by parts}\\
&= x\sin(\ln x) - \int x\left(\cos(\ln x)\right)(\ln x)'\,\mathrm{d}x\\
&= x\sin(\ln x) - \int \cos(\ln x)\mathrm{d}x && \text{int. by parts}\\
&= x\sin(\ln x) - \left(x\cos(\ln x) - \int x\mathrm{d}(\cos(\ln x))\right)\\
&= x\sin(\ln x) - x\cos(\ln x) + \int x(-\sin(\ln x))(\ln x)'\mathrm{d}x\\
&= x\sin(\ln x) - x\cos(\ln x) - \int \sin(\ln x)\mathrm{d}x && \text{add } \textstyle\int \sin(\ln x)\mathrm{d}x \text{ to both sides}\\
2\int \sin(\ln x)\mathrm{d}x &= x\sin(\ln x) - x\cos(\ln x)\\
\int \sin(\ln x)\mathrm{d}x &= \frac{x}{2}\left(\sin(\ln x) - \cos(\ln x)\right) \quad .
\end{aligned}
$$

Solution. 7.s

$$
\int \ln x\mathrm{d}x = x\ln x - \int x\mathrm{d}(\ln x) = x\ln x - \int \frac{x}{x}\mathrm{d}x = x\ln x - x + C \quad .
$$

Solution. 7.u

$$
\begin{aligned}
\int \frac{\ln x}{\sqrt{x}}\mathrm{d}x &= \int (\ln x)2\mathrm{d}\left(\sqrt{x}\right) && \text{integrate by parts}\\
&= (\ln x)2\sqrt{x} - \int 2\sqrt{x}\mathrm{d}(\ln x)\\
&= 2\sqrt{x}\ln x - 2\int \frac{\sqrt{x}}{x}\mathrm{d}x\\
&= 2\sqrt{x}\ln x - 2\int x^{-\frac{1}{2}}\mathrm{d}x\\
&= 2\sqrt{x}\ln x - 4\sqrt{x} + C\\
&= 2\sqrt{x}(\ln x - 2) + C \quad .
\end{aligned}
$$

8. Compute $\int x^n e^x \mathrm{d}x$, where n is a non-negative integer.

Solution. 8

$$
\begin{aligned}
\int x^n e^x\mathrm{d}x &= \int x^n \mathrm{d}e^x\\
&= x^n e^x - \int e^x \mathrm{d}x^n\\
&= x^n e^x - n\int x^{n-1}e^x\mathrm{d}x\\
&= x^n e^x - n\left(\int x^{n-1}\mathrm{d}e^x\right)\\
&= x^n e^x - n\left(x^{n-1}e^x - \int (n-1)x^{n-2}e^x\mathrm{d}x\right)\\
&= x^n e^x - nx^{n-1}e^x + n(n-1)\int x^{n-2}e^x\mathrm{d}x\\
&= \ldots \text{(continue above process)} \ldots\\
&= x^n e^x - nx^{n-1}e^x + n(n-1)x^{n-2}e^x + \ldots\\
&\quad +(-1)^k n(n-1)(n-2)\ldots(n-k+1)x^{n-k}e^x\\
&\quad +\cdots+(-1)^n n!e^x + C\\
&= C + \sum_{k=0}^{n}(-1)^n \frac{n!}{(n-k)!}x^{n-k}e^x \quad .
\end{aligned}
$$